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# Explain-to-Learn Microlearning for Metacognitive Regulation in Academic Speaking

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**ABSTRACT**

This study examined whether explain-to-learn microlearning could strengthen metacognitive regulation and academic speaking performance in higher education. Over eight weeks, 80 Indonesian undergraduates completed short multimedia modules, oral explanation tasks, and structured reflection prompts. Using a classroom-based pre-post evaluation informed by design-based research, the study analyzed speaking scores, metacognitive regulation, and learner perceptions. Three findings emerged: 1) metacognitive regulation improved significantly across all three dimensions, with the largest gains in evaluation and planning (partial $\eta^2$ = .59); 2) academic speaking performance increased substantially across all dimensions, with the strongest gains in idea organization and clarity of explanation ($d$ = 1.01); and 3) students rated the design highly for usefulness, cognitive engagement, and instructional clarity. The findings suggest that microlearning becomes more instructionally useful for academic speaking when brief content delivery is followed by learner explanation and explicit metacognitive scaffolding.



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## 1. INTRODUCTION

University students are expected to do more than speak fluently. They must explain concepts, organize disciplinary arguments, and make their reasoning clear to listeners who are also learning. This is a cognitively demanding task, and Goh [3] is right to argue that speaking development depends not only on what students produce, but on the internal processes (planning what to say, monitoring whether an explanation is landing, evaluating what fell short) that regulate performance before, during, and after speech. In most university speaking classrooms, however, those processes are left largely implicit. Instruction tends to prioritize observable outcomes: fluency, accuracy, pronunciation. The metacognitive work that supports them rarely gets a dedicated place in the curriculum.

In Indonesia, the stakes are higher than they might first appear. National standards for higher education now require curricula to be organized around graduate learning outcomes that explicitly include effective communication, and they mandate student-centered methods such as discussion, seminar, and project-based learning as pathways to those outcomes [1]. The policy intent is clear. In practice, though, university students frequently struggle with precisely these performances: local studies report persistent weaknesses in scientific presentation, limited oral proficiency, and public speaking anxiety among undergraduates [2]. The demand for communicative competence has moved faster than the instructional infrastructure for building it.

This is not simply a matter of exposure or practice hours. Academic speaking is cognitively heavier than everyday conversation because it requires learners to simultaneously manage content organization, self-monitoring, and audience awareness under real-time pressure. Flavell [4] identified planning, monitoring, and evaluation as foundational regulatory processes in learning; Zimmerman [5] extended that framework to self-regulated performance more broadly. Applied to speaking, these processes matter: Paterson [6] showed that structured metacognitive reflection can drive meaningful speaking improvement, and Zhang et al. [7] found consistent associations among metacognitive strategy use, task demand, and L2 speaking quality. Safari and Fitriati [8] similarly observed that higher-performing speakers are distinguishable from weaker ones partly through how they regulate their own production. The issue is not whether metacognition matters for speaking; the evidence suggests it does. The issue is that metacognitive support in university courses is typically incidental, delayed, or invisible.

Microlearning has attracted attention as one way to make instruction more focused and cognitively manageable. Short, targeted modules can reduce complexity, lower extraneous load, and keep learners engaged across constrained time windows [9], [10], [11], [12]. There is also emerging evidence that microlearning can support speaking development when it is embedded in meaningful communicative activity rather than used as isolated drill [13].

But brevity is not a pedagogy. When short modules function mainly as segmented content delivery, they can improve access and convenience without requiring learners to do anything generative with what they receive: reorganize it, express it aloud, or evaluate whether their explanation actually communicated what they intended. The microlearning literature has not yet worked through this problem, particularly not in the context of academic speaking where exactly this kind of elaborative, self-regulated processing matters most.

Explain-to-learn strategies offer a way to close that gap. Fiorella and Mayer [14] describe explanation as a generative learning activity, one that requires learners to select, organize, and integrate information rather than simply retrieve it. Lachner et al. [15] and Bai et al. [16] show that embedding explanation prompts in multimedia learning environments can deepen processing and surface gaps in understanding. Mayer's cognitive theory of multimedia learning [17] adds that brief, well-structured input can reduce extraneous burden and free cognitive resources for exactly this kind of active knowledge construction. The instructional implication is that microlearning might become more powerful when it is not the end of a learning cycle, but the beginning of one: short input followed by learner explanation followed by explicit reflection on how well that explanation worked.

This study tested that design in a classroom context. It examined a three-component instructional cycle comprising brief multimedia input, oral explanation tasks, and structured metacognitive prompts for planning, monitoring, and evaluation, implemented over eight weeks with undergraduate students in an Indonesian university. Three questions guided the inquiry: 1) to what extent does the intervention influence students' metacognitive regulation in academic speaking; 2) how does it affect academic speaking performance across key dimensions of speaking quality; and 3) how do students perceive the usefulness, cognitive engagement, and clarity of the instructional design?

## 2. METHOD

This study used a classroom-based pre-post evaluation informed by a design-based research orientation [18], [19]. Eighty undergraduate students enrolled in a language education program at a private university in Cimahi, West Java, participated through their intact classes. They had prior exposure to academic speaking coursework but varied in proficiency and metacognitive awareness. Participation followed standard institutional procedures for consent, anonymity, and non-evaluative data use.

The intervention ran for eight weeks within a web-based microlearning environment, following a repeated instructional cycle across all sessions:

Brief multimedia input (short video, annotated example, or infographic) → Oral explanation task (recorded in own words) → Structured reflection prompts (planning · monitoring · evaluation)

The design moved through a standard pre-post arc: Pre-test → Intervention (8 weeks, repeated cycles) → **Post-test**

Each cycle was structured so that microlearning did not stop at content delivery. Students had to re-explain what they had just encountered in their own spoken words, then reflect explicitly on how well they had communicated it, connecting brief input to generative processing and self-regulated review.

Data came from three sources. Speaking performance was assessed through task-based oral presentations scored on idea organization, clarity of explanation, strategic delivery, and responsiveness; two independent raters used a five-point rubric. Metacognitive regulation was measured with a 15-item questionnaire covering planning, monitoring, and evaluation. Learner perceptions were gathered via an eight-item survey on usefulness, cognitive engagement, and instructional clarity, supplemented by open-ended responses.

All instruments were reviewed by subject-matter experts and piloted before use. Repeated-measures MANOVA was used for metacognitive regulation and paired-sample t-tests for speaking performance, with partial $\eta^2$ and Cohen's d as effect size indices. Thematic analysis of open-ended responses was used to interpret, not replace, the quantitative findings [20].

## 3. RESULTS AND DISCUSSION

Three findings emerged from the pre-post analysis, one for each of the study's three research questions. The pattern across all three was directionally consistent, but the distribution of gains within each outcome tells a more specific story about how the intervention worked.

### 3.1 Metacognitive regulation gains

All three dimensions of metacognitive regulation (planning, monitoring, and evaluation) improved from pre-test to post-test, with the overall multivariate effect reaching significance at Wilks' $\Lambda = .41$, $F(3, 76) = 44.62$, $p < .001$, partial $\eta^2 = .59$. The gains were not evenly distributed across the three dimensions, and that unevenness is where the instructional story sits

**Table 1** The descriptive pattern.

| VARIABLE | PRE-TEST M (SD) | POST-TEST M (SD) | MEAN DIFFERENCE |
|---|---|---|---|
| Planning | 3.12 (0.78) | 4.17 (0.62) | 1.05 |
| Monitoring | 3.45 (0.71) | 4.02 (0.68) | 0.57 |
| Evaluation | 3.08 (0.82) | 4.26 (0.59) | 1.18 |
| Overall speaking performance | 2.89 (0.74) | 3.94 (0.65) | 1.05 |

Planning and evaluation improved substantially (partial $\eta^2 = .39$ and $.42$, respectively), while monitoring improved more modestly (partial $\eta^2 = .22$). This gap reflects something real about how speaking regulation responds to instruction. Monitoring is difficult to scaffold directly: it requires students to track their own production under the real-time pressure of ongoing speech. Planning and evaluation, by contrast, happen before and after speech, which is precisely where the intervention's explicit prompts applied force. The result is consistent with what Paterson [6] and Zhang et al. [7] observed: that reflection-oriented support tends to produce stronger instructional effects than attempts to improve regulatory control during live speech production.

### 3.2 Academic speaking improvement

Speaking performance showed a large overall gain from pre-test to post-test ($t(79) = 9.87$, $p < .001$, **d** = 1.01), but the aggregate effect size flattens what is a differentiated pattern across the four speaking dimensions.. Table 2 summarizes these results.

**Table 2.** Dimension of speaking

| DIMENSION | T(95) | P-VALUE | COHEN'S D |
|---|---|---|---|
| Idea organization | 10.12 | < .001 | 1.15 |
| Clarity of explanation | 9.56 | < .001 | 0.98 |
| Strategic delivery | 8.43 | < .001 | 0.87 |
| Responsiveness | 7.89 | < .001 | 0.82 |
| Overall performance | 9.87 | < .001 | 1.01 |

The strongest improvements fell in idea organization (**d** = 1.15) and clarity of explanation (**d** = 0.98), the two dimensions most directly tied to what the intervention required students to do. When learners repeatedly re-express disciplinary content in their own words, they are forced to decide what to foreground, what to cut, and whether their framing is actually making sense to a listener. That cycle of selection and reformulation maps almost exactly onto what these two dimensions assess. Explanation-centered and video-supported speaking research supports this reading [21], [22], as does technology-mediated work showing that digital speaking tasks tend to improve proficiency when they require meaningful output rather than passive media consumption [23], [24].

The comparatively smaller gain in responsiveness (**d** = 0.82) points to a genuine boundary of the design, not a flaw in execution but a constraint of scope. The intervention prioritized individual explanation tasks; dialogic exchange, where students would need to adapt their speech in real time to a live listener, was largely absent from the cycle. The data reflect that focus. Review-level evidence consistently shows that microlearning gains depend on how units are embedded in broader activity rather than on brevity alone [9], [10], [12], [25], [26], and monologue-oriented tasks are better suited to organizational control than to spontaneous interaction [27].

Correlation between speaking performance and each metacognitive dimension (planning, **r** = .68; monitoring, **r** = .54; evaluation, **r** = .71; all $p < .01$) suggests that regulatory improvement and speaking improvement moved together rather than independently. These are associations, not causal evidence, and the design does not support a mediation claim. The pattern is taken up further in the integrated interpretation below.

### 3.3. Student perceptions of the design

Beyond what the performance and regulation data showed, students were also asked whether the design actually felt useful to them. Perceived usefulness averaged M = 4.23 (SD = 0.71), cognitive engagement M = 4.15 (SD = 0.76), and instructional clarity M = 4.08 (SD = 0.79). The ratings were uniformly high, but the open-ended responses are more informative about what drove them.

Three themes recurred across the dataset: short modules felt manageable rather than overwhelming; the oral explanation task surfaced gaps in understanding that students had not noticed during passive viewing; and the reflection prompts helped them approach subsequent speaking tasks more deliberately. What is notable is the specificity. Students were not describing general satisfaction with a digital platform; they were identifying the structural features of the design in terms that closely matched the design's own theoretical rationale. That alignment between perceived mechanism and intended mechanism gives the quantitative gains a clearer interpretive grounding.

### 3.4 Integrated interpretation

Reading the three sets of findings alongside each other sharpens the picture. Metacognitive gains were largest in the dimensions tied to before and after speaking: planning and evaluation. Speaking gains were largest in the dimensions most dependent on knowing what to say and how to frame it: organization and explanatory clarity. These patterns mirror each other structurally, and the correlation between evaluation and speaking performance (**r** = .71) suggests they were not independent: students who improved most in post-task regulatory review were also those who showed the strongest gains in how they organized and delivered their explanations. The intervention likely worked not by delivering content more efficiently, but by forcing a repeated cycle of reorganization, expression, and reflection, with each pass giving students a clearer internal model of what a well-explained idea actually sounds like.

That interpretive claim has to be stated within honest limits. The study ran in a single institutional context, relied on self-report for metacognitive regulation, and had no comparison group. Without a control condition it is not possible to rule out maturation, concurrent course effects, or other influences on performance; the correlational associations do not establish mediation. What the study can defensibly claim is that substantial gains in both outcomes appeared under conditions where microlearning was coupled with generative explanation and explicit metacognitive scaffolding, and that students described the mechanism in terms consistent with how the design was theorized. That is a meaningful but bounded finding. Whether this design outperforms alternative instructional arrangements, and whether gains hold across different populations and longer timeframes, are questions that need more controlled and sustained work.

## 4. CONCLUSION

The gains in this study were real, but the more useful finding is the pattern within them. Planning and evaluation improved more than monitoring; idea organization and explanatory clarity improved more than responsiveness. That is not a random distribution; it tracks the structure of the intervention almost exactly. The design applied deliberate pressure to pre-task organization, post-task review, and the specific demands of formulating and expressing a coherent explanation aloud. The dimensions that moved most were the ones the intervention was built to target.

The practical implication is narrower than a general endorsement of microlearning for speaking. Short content units alone are unlikely to produce this kind of gain. What appears to matter is what learners are required to do immediately after receiving input: whether they must reorganize it, express it in their own spoken words, and then evaluate how well their explanation worked. Under those conditions, microlearning stops functioning as a delivery format and becomes the first step in a generative cycle. That is the design claim this study advances, and it is a testable one.

The single-context, no-control design limits how far that claim can currently travel. Replication in more varied settings, with longer durations, more interactive speaking tasks, and a comparison condition, is the clear next step, one that would allow stronger statements about relative effectiveness rather than pre-post change alone. For now, the evidence is sufficient to give university speaking instructors a structured and theoretically grounded model: one that takes the communicative competences that Indonesian national curriculum standards now formally require [1] and offers a concrete instructional path for building them, not just a policy mandate to meet.


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Restu Bias Primandhika contributed to conceptualization, methodology, formal analysis, investigation, data curation, writing the original draft, visualization, project administration, and funding acquisition. Nani Solihati contributed to conceptualization, methodology, validation, writing review and editing, supervision, and project administration. Siti Zulaiha contributed to conceptualization, methodology, validation, writing review and editing, supervision, and project administration. While Aurelia Sakti Yani contributed to data collection, curation, and project administration.

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| C : **C**onceptualization | I : **I**nvestigation | Vi : **Vi**sualization |
| M : **M**ethodology | R : **R**esources | Su : **Su**pervision |
| So : **So**ftware | D : **D**ata Curation | P : **P**roject administration |
| Va : **Va**lidation | O : Writing - **O**riginal Draft | Fu : **Fu**nding acquisition |
| Fo : **Fo**rmal analysis | E : Writing - Review & **E**diting | |

## CONFLICT OF INTEREST STATEMENT

The authors state that they have no conflict of interest.

## INFORMED CONSENT

All student participants were provided with a written informed consent form prior to their involvement in the study. By signing the form, they confirmed their voluntary participation and acknowledged that their learning activities would be observed and documented as part of the research process.

## ETHICAL APPROVAL

This study involved university students and was conducted in accordance with relevant institutional guidelines for educational research, including voluntary participation, anonymity, confidentiality, and non-evaluative participation. No separate ethics approval number was reported in the study records used for this manuscript.

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available because they contain information that could compromise the privacy of research participants.

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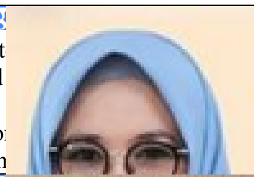
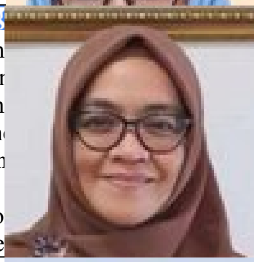
|  | **Nani Solihati** ndonesian Language and Literature Education<br>at the Post-Graduat hammadiyah Prof. DR. Hamka (UHAMKA).<br>She has researched ely published in the areas of critical thinking<br>tasks in language hology, gender inequality in literature, and<br>character educatio games. She can be contacted at email:<br>nani_solihati@uhan |
|  | **Siti Zulaiha** Professor, the Head of the Master's Degree<br>Program in English ecturer at the Graduate School of Universitas<br>Muhammadiyah Pr MKA). She conducts research on primary<br>school foreign lan , and assessment, implementation of next-<br>generation assessm culum development, and the perceptions and<br>pedagogical challen s in language classrooms. These studies have<br>been published in book chapters, and academic publications,<br>including insights o l English linguistics.<br>She can be contacte hamka.ac.id. |
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